

Gibonacci number

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Description

In mathematical terms, the normal sequence **F(n)** of Fibonacci numbers is defined by the recurrence relation

$$F(n) = F(n-1) + F(n-2)$$

with seed values

$$F(0) = 1, F(1) = 1$$

In this **Gibonacci numbers** problem, the sequence **G(n)** is defined similar

$$G(n) = G(n-1) + G(n-2)$$

with the seed value for **G(0)** is 1 for any case, and the seed value for **G(1)** is a random integer **t**, (**t** ≥ 1). Given the i-th Gibonacci number value **G(i)**, and the number **j**, your task is to output the value for **G(j)**

Input

There are multiple test cases. The first line of input is an integer **T** < 10000 indicating the number of test cases. Each test case contains 3 integers **i**, **G(i)** and **j**. $1 \leq i, j \leq 20$, $G(i) < 1000000$

Output

For each test case, output the value for **G(j)**. If there is no suitable value for **t**, output -1.

Samples

input

Copy

```
4
1 1 2
3 5 4
3 4 6
12 17801 19
```

output

Copy

```
2
8
-1
516847
```

Source

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